

SUMMARY	Computer Scientist working in Machine Learning with 10+ years of research experience.		
INTERESTS	Machine Learning, Deep Learning, Natural Language Processing		
EDUCATION	Northwestern University , Evanston, Illinois USA		
	Ph.D., Computer Science		Sep '19
	Master of Science, Computer Science		
	Birla Institute of Technology & Science , Pilani, Rajasthan India		
PROGRAMMING SKILLS	Master of Engineering, Software Systems		May '12
	Bachelor of Engineering, Chemical Engineering		
	Python, Pandas, Scikit-Learn, XGBoost, Streamlit, Dash, Tensorflow, Google Cloud Libraries		
PROFESSIONAL EXPERIENCE	<i>Lead Data Scientist (ML Research Team)</i> , American Family Insurance , Boston		Oct'24-
	- Generative AI-based Conversational Intelligence for QA : - Designed effective prompts for Gemini Multimodal models to assess claim quality, leveraging its multimodal capabilities for comprehensive analysis. - Collaborated with the Claims QA team to optimize auditing processes and questionnaires, streamlining workflows and improving efficiency. - Leadership: - Host of the AmFam Fairness & Ethics in ML Seminar. - Collaborated closely with UW-Madison professors as part of the Amfam Data Science Institute, leading to two publications.		
	<i>Senior Data Scientist (ML Research Team)</i> , American Family Insurance , Boston		Feb'24-Oct'24
	- Generative AI for Audio Call Summarization & Entity Extraction : - Contributed to Generative AI pipelines for audio call summarization and entity extraction, ensuring robust performance across formats. - Investigated Whisper using pyannote for speaker identification and used models like Gemini-1.5-Pro, Mistral-Au-Large, and GPT-4-Turbo for summarization. - Custom Generative AI Chatbot Development: - Implemented a custom Generative AI chatbot leveraging technologies like streamlit, langchain, and GPT-4-Turbo/Claude-Sonnet, enhancing customer experience.		
	<i>Data Scientist (ML Research Team)</i> , American Family Insurance , Madison & Boston		Oct'19-Feb'24
	- User-Based Insurance (in collaboration with major US automaker): - Developed generalized linear and additive models based on telematics features for auto insurance in collaboration with a major US automaker. - Claims-Channeling System: - Co-designed a claims channeling system utilizing transfer learning and performed an ablation study to enhance model accuracy for routing claims to experts. - Financial Forecasting: - Developed long and mid-term financial forecasts using an ensemble ESRNN+SARIMA+GBM model, deploying a financial dashboard for leadership. - Motor Vehicle Violation: - Built an ML decision system to predict motor vehicle violation risk using internal and external data. - Leadership/Outreach: - Mentored rotational associate data scientists. - Participated as a panelist for company-wide data privacy discussions on fairness, privacy, and bias.		
	<i>Data Science Intern</i> , Northwestern Mutual , Milwaukee, Wisconsin		Jun - Aug '18
	Developed distributed OCR algorithms for detecting responses from scanned questionnaires		
	<i>Research Intern</i> , Boeing Cybersecurity , Sunnyvale, California		Jun - Sep '13
	Developed machine learning system using generated synthetic user profiles with different demographic and interest features for analyzing ads across profiles		

- Chemical Property Prediction using Molecular Fingerprints (Tensorflow, Scikit Learn)
 - ◊ Developed a multi-input input neural network architecture by merging different molecular representations (SMILES and fingerprints) for predicting chemical properties and reduced the mean absolute error by half compared to state-of-the-art architectures (CheMixNet architecture)
 - ◊ Designed Bagged Ensemble models for predicting power conversion efficiency of solar cells using chemical fingerprints, and achieved mean absolute percentage error between 1.5-2 %
 - ◊ Developed a transfer learning solution to predict solar cell properties with mean absolute percentage error below 1 % (SINet architecture)
- Developed Predictive Model for Additive Manufacturing (Tensorflow, Keras)
 - ◊ Created time series models for temporal analysis of temperature and heat flux data
 - ◊ Investigated Recurrent Neural Network models to predict point-wise temperature information for accelerating additive manufacturing simulations
 - ◊ Developed an iterative real-time predictive model using bagged decision trees
- Classification of Anonymous Posts using Recurrent Neural Networks (Keras, Scikit Learn)
 - ◊ Developed customized vector model using crowd-sourced (Urban Dictionary) & psycho-lingual (LIWC) dictionaries
 - ◊ Explored Word2vec, GloVe and FastText embedding schemes (Gensim)
 - ◊ Attained prediction accuracy of 79.8 % and 78.1 % using ensemble and LSTM models respectively

F. Li, Q. Zhu, K. Yuan, F. Ji, **A. Paul**, P. Lee, V.C. Radeloff, M. Chen. "**Projecting Large Fires in the Western US with a More Trustworthy Machine Learning Method**", *Earth's Future*, 2024

Y. Mao, M. Hasan, **A. Paul**, V. Gupta, K. Choudhary, F.M. Tavazza, W. Liao, A. Choudhary, P. Acar and A. Agrawal. "**An End-to-End AI-Driven Microstructure Optimization Framework for Elastic Properties of Titanium Beyond Cubic Crystal Systems**", *Nature Partner Journal on Computational Materials*, 2023

A.Dimri, **A.Paul**, D.Girish, P.Lee, S.Afra and A.Jakubowski. "**A Multi-input Multi-label Claims Channeling System Using Insurance-Based Language Models**", *Expert Systems With Applications*, 2022

K.Ness, **A. Paul**, L. Sun and Z. Zhang. "**Towards a generic physics-based machine learning model for geometry invariant thermal history prediction in additive manufacturing**", *Journal of Materials Processing Technology*, 2022 (*Special Issue on AI in Advanced Manufacturing*)

R.Richards, and **A. Paul**. "**An Attention-driven LSTM Network for High Throughput Virtual Screening of Organic Photovoltaic Candidate Molecules**", *Solar Energy*, 2021

Z.Yang, D. Jha, **A. Paul**, W. Liao, A. Choudhary and A. Agrawal. "**A General Framework Combining Generative Adversarial Networks and Mixture Density Networks for Inverse Modeling in Microstructural Materials Design**", *NeurIPS Workshop on Machine Learning for Engineering Modeling, Simulation and Design*, 2020

D.Jha, L.Ward, **A. Paul**, W. Liao, A. Agrawal, A. Choudhary and C. Wolverton. "**ElemNet: Deep Learning the Chemistry of Materials From Only Elemental Composition**", *Nature Scientific Reports*, 2018

A. Paul, M.Mozaffar, Z. Yang, W. Liao, A. Choudhary, J.Cao and A. Agrawal. "**A real-time iterative approach for temperature profile prediction in additive manufacturing processes**", *6th IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, 2018

A. Paul, D.Jha, R. Al-Bahrani, W. Liao, A. Choudhary and A. Agrawal. "**Transfer Learning Using Ensemble Neural Nets for Organic Solar Cell Screening**", *International Joint Conference on Neural Networks*, 2019

A. Paul, D.Jha, R. Al-Bahrani, W. Liao, A. Choudhary and A. Agrawal. "**CheMixNet: Mixed DNN Architectures for Predicting Chemical Properties using Multiple Molecular Representations**", *NeurIPS Workshop on Machine Learning for Molecules and Materials*, 2018

M.Mozaffar, **A. Paul**, R. Al-Bahrani, S. Wolff, A. Choudhary, A. Agrawal, K. Ehmann and J.Cao. "**Data-Driven Prediction of the High-Dimensional Thermal History in Directed Energy Deposition Processes via Recurrent Neural Networks**", *Manufacturing Letters*, 2018

J.Birnholz, N.A.R. Merola, and **A. Paul**. "**Is it Weird to Still Be a Virgin?: Anonymous, Locally Targeted Questions on Facebook Confession Boards**", *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems. ACM*, 2015.